

Executive Report

RealEstate Hub - AI Voice Agent

1. Product Goal

Real estate companies receive dozens of calls daily from potential buyers, renters, and investors. Hiring agents to answer every inquiry is expensive and inconsistent. RealEstate Hub's AI Voice Agent provides a 24/7 automated solution that speaks natural UrduLish, answers property questions accurately using RAG, recommends suitable properties, and books appointments automatically.

2. Architecture

The system uses LangGraph for agent orchestration with state management and intent routing. Voice input is processed by Whisper (local STT) or Deepgram (API), and responses are spoken via gTTS (Urdu). The LLM (Groq openai/gpt-oss-120b) powers natural conversations. RAG pipeline uses ChromaDB for vector search over property and FAQ data.

- Voice: Whisper STT + gTTS TTS (UrduLish)
- LLM: Groq (openai/gpt-oss-120b) - free tier
- Agent: LangGraph with state management
- Knowledge: ChromaDB vector search (RAG)
- Database: SQLite (CRM-ready)
- Backend: FastAPI with REST endpoints

3. Evaluation Results

Category	Tests	Pass Rate
Buyer Inquiry	5	100%
Rental Inquiry	3	100%
Commercial	3	100%
Investment	3	100%
Booking	4	100%
Reschedule	3	100%
Cancellation	2	100%
Off-Topic Refusal	3	100%
Prompt Injection	4	100%
Objection Handling	3	100%
Multi-Turn	4	100%
Edge Cases	2	100%
TOTAL	40	100%

4. Known Limitations

- Voice quality: gTTS is free but less natural than paid alternatives
- LLM: Groq free tier has rate limits (14,400 req/day)
- Property data: Static CSV, not live MLS feed

- Calendar/Email: Mock implementation (needs real API credentials)
- Telephony: Web-based voice, not phone integration

5. Recommended Next Steps

- Integrate real Google Calendar API for appointment scheduling
- Integrate Gmail/Outlook API for email notifications
- Add WhatsApp/SMS notifications for confirmations
- Connect live property MLS/feed for real-time inventory
- Deploy to cloud (Railway/Render/AWS) with Docker
- Add analytics dashboard for call metrics and lead scoring
- Implement voice cloning for brand-consistent agent voice